GROUP 11

EMMANUEL LIMO DSE-01-8840/2024

GRACE WAIRIMU DCF-01-0109/2024

ERIC OWUOR DSE-01-8673/2024

MOSES MUIRURI DSE -01-0103/2025

SHERRY ADHIAMBO DCF-01-0046/2025

HAGONO LAVENDER DCF-01-0157/2024

JOSPHAT KARIUKI DSE-01-8682/2024

ADVANTAGES OF VIRTUALIZATION IN CYBER SECURITY

Safe testing environment – this is also known as sandboxing .Cyber security experts use vms when running suspicious programs ,opening untrusted files and when testing the capabilities of known malwares without risking the security of their systems; Incase the vm is compromised it can always be deleted or be reset.

Easy snapshot and roll back -sometimes these virtual machines are used to run the real systems .virtual machines offer snapshots features which allows a rollback ,this makes it easier to regain stability in case the system is attacked .Cyber experts could easily restore the system to its stable state.

Improved network security testing-Virtual machines help experts to build virtual network infrastructure to simulate real network system enabling secure testing of firewalls, attacks, and intrusion detection systems.

Cost effective-Instead of buying multiple hardware system experts use one device to create multiple virtual machines hence saving power and capital. This also reduces the maintenance and eases the upgrade since all this must be done on less hardware machines.

Rapid deployment -Vms makes it possible to create servers, honey pots and test environment in brief time compared to setting up real hardware components.

VIRTUALIZATION OF NETWORK DEVICES AND SERVICES

Virtualization of network devices and services is the process of creating **software-based versions** of routers, switches, firewalls, servers, and security tools instead of using physical hardware, for example cyber ops workstation for cyber security. This is achieved through virtual machines. A virtual machine (VM) works by using a **hypervisor**, which is software that allows one physical computer to run many separate virtual computers at the same time.
The hypervisor divides the real hardware (CPU, RAM, storage, network) and gives each VM its own share.
Inside the VM, an operating system runs just like it would on a real computer.
Because each VM is isolated, whatever happens inside it—malware, errors, testing—does not affect the host machine or other VMs.

When creating a virtual machine, the following procedures are followed.

Install a hypervisor.

Create a new virtual machine.

Allocate CPU, RAM, and storage.

Install an operating system in the VM.

Configure virtual network interfaces.

DATA FORMATS USED IN NETWORK AUTOMATION

JavaScript object notation-This is a light weight and text-based data format used in storing and exchanging data between automation tools and network devices. It is easy to read and understand. It is used by APIs to return configuration data.

Yaml ain't markup language (yaml)-it is used in configuration files it human readable and simple. It is clean and readable since it uses indentation instead of brackets. Used by ansible (*a YAML file that defines a series of automated tasks to be executed on IT infrastructure*)

XML Extended markup language .XML is a **structured document format** that uses tags to describe data.
It is commonly used by older network devices and protocols like **NETCONF** (*a standard protocol for installing, manipulating, and deleting the configuration of network devices*)for sending configuration and monitoring data.

API AVAILABLE FOR CYBER SECURITY EXPERTS

Metasploit RPC -Allows experts to control Metasploit through scripts for Penetration testing.

Exploit automation and Payload delivery tests.

Virus Total API – checks files and URLs for malware using multiple antivirus engines.

Shodan API – scans the internet for exposed devices, open ports, and vulnerabilities.

Nmap API - Helps automate recon for both defense and penetration testing. Used to automate tasks like port, OS, and service scanning.

DVR (Digital Video Recorder) and NVR (Network Video Recorder)

DVR (Digital Video Recorder)

A DVR is a device used in CCTV systems that records video from analog cameras.
The cameras connect using coaxial cables, and the DVR converts the analog signal into digital format for storage.

Works with analog cameras.

Uses coaxial cables.

Video is processed inside the DVR.

Used in older or lower-cost CCTV systems.

NVR (Network Video Recorder)

An NVR is a device that **records video from IP cameras**.
The cameras send digital video over a **network (Ethernet)**, and the NVR stores it directly.

Key points:

Works with **IP (network) cameras.**

Uses **Ethernet cables (LAN)**

Video is **processed in the camera** itself.

Used in modern, high-quality CCTV systems.